THE ATMOSPHERE 2

11

The Noble gases In 1904 Sir William Ramsay was awarded the Nobel Prize in Chemistry for discovering a new Group of elements, the Noble gases. This Group is now known to consist of helium (He), neon (Ne), argon (Ar), krypton (Kr), xenon (Xe) and radon (Rn). Radon was discovered by Fredrich Ernst Dorn in 1898. These gases are all colourless, tasteless and very unreactive. They are unreactive because they have full outer (valence) electron shells. In total, the Noble gases make up around 1% of the earth's atmosphere (by volume); argon is by far the most common. Although helium is estimated to make up around 23% of the material in the Universe, it only makes up about 0.0005% of Earth's atmosphere.

Helium (from the Greek word *Helios*, meaning Sun) is used in cryogenics because it can (when liquid) reach temperatures very close to absolute zero (–273.15°C). Cryogenics is the science that deals with the behaviour of materials at very low temperatures (*cryo*: icy; *genic*: produced by nature). Another use of helium is in lighter-than-air balloons.

When **neon** (from the Greek *neos*, meaning new) gas has an electric current passed through it while under low pressure, the gas emits a reddish light. This makes neon useful in making advertising signs and also in some lasers.

Argon (from the Greek *argon* meaning lazy) can be used when welding reactive metals to provide an inert atmosphere to stop the metals from reacting with oxygen. Light globes can also contain argon. In these globes a thin tungsten (W) wire glows white-hot when an electric current is passed through it. The inside of the globe is filled with argon.

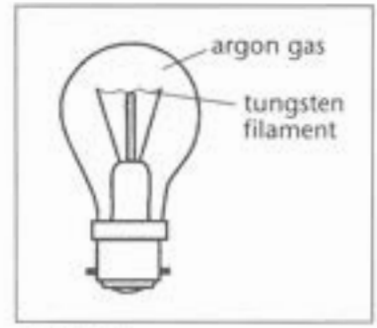


Figure 1

1 Give a reason why the gas argon is used instead of air in the light globe shown in figure 1.

Krypton (from the Greek *kryptos*, meaning hidden) forms some compounds with oxygen and chlorine. It is also used in lighting. **Xenon** (from the Greek *xenon*, meaning stranger) is the most reactive of the Noble gases. It is used in powerful lamps and radiation detectors. **Radon** is a radioactive gas that is produced by the decay of radium (Ra). Radon can be used to treat cancers (radiotherapy) and in taking X-ray type pictures of the human body (radiography).

2 The noble gases are all monatomic. Explain what is meant by *monatomic*.

3 The noble gases can be extracted from liquid air by *fractional distillation*. What property of the gases allows fractional distillation to occur?

4 Draw up a table to summarise the information about the noble gases. Use the headings shown below.

Name	Symbol	Use	Origin of name

5 What is *cryogenics*?

6 Give the symbol for tungsten.

7 Give the symbol for radium.

8 Why are the Noble gases so unreactive?

Carbon dioxide This gas is naturally present in the atmosphere, but levels have been increasing since humans began burning fossil fuels (coal, oil and natural gas) in large quantities. It is a significant greenhouse gas that contributes to global warming. Many people are concerned about the possible consequences of increasing global warming. Carbon dioxide (CO_2) turns limewater (calcium hydroxide) solution, $Ca(OH)_2$ (aq), 'milky' forming insoluble calcium carbonate ($CaCO_3$). This reaction is used as a test for carbon dioxide gas.

9 Write a balanced formula equation for the reaction between carbon dioxide and limewater.

10 Pollution can be defined as: *The contamination of an environment, especially with waste made by humans*. Do you consider carbon dioxide a pollutant? Explain your answer.

11 Mark each statement as true or false.

a The Noble gases form a large proportion of Earth's atmospheric gases.

b Carbon dioxide would not be present in the atmosphere if humans hadn't polluted it.

c Oxygen turns lime water from clear to milky.

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Pollution in the atmosphere

There are substances in the atmosphere that we think are harmful to both the environment and people. These substances are often grouped under the title **pollution**. Sources of pollution include motor vehicles, industry, power-generating stations and human homes. The petrol (internal combustion) engine has the following theoretical reaction:

hydrocarbon fuel + oxygen → carbon dioxide + water (plus heat) (in excess oxygen)

Octane (C_8H_{18}) is a liquid hydrocarbon often used as a benchmark for judging other fuels (petrol).

12 Define *pollution*.

13 Write a balanced formula equation for the combustion of octane in a plentiful supply of air.

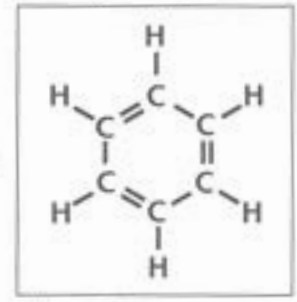


Figure 2

Unfortunately, in practice, the reaction of hydrocarbon fuel and oxygen (even with a pure hydrocarbon fuel) doesn't quite follow this path of forming the products carbon dioxide and water. Other products include unburned hydrocarbons, carbon monoxide and nitrogen oxides. Sulfur dioxide and particulate matter (small particles that can pass into the lungs) are also produced by diesel engines.

Hydrocarbons One hydrocarbon that causes special concern is benzene (see figure 2). Petrol and diesel fuel contain only about 1% benzene, but it is toxic and carcinogenic (cancer-forming).

14 What is the molecular formula for benzene?

15 Apart from exhaust fumes, how can hydrocarbon fuels be released into the atmosphere?

16 Give two reasons why contact with benzene should be minimised.

Carbon monoxide (CO) Wood stoves, cigarette smoke and unvented gas heaters are also sources of carbon monoxide. Carbon monoxide is a colourless, odourless and poisonous gas.

17 Explain why very high levels of carbon monoxide occur in areas of traffic congestion.

Nitrogen oxides NOx is often used as a shorthand way of representing the various oxides of nitrogen. NOx is produced by vehicles and power stations. It is harmful to breathing systems and combines with unburned hydrocarbons to form ground-level ozone (O_3) and smog. It is also a major contributor to acid rain.

18 Nitrogen dioxide (NO_2) can dissolve in water to form dilute nitric acid (HNO_3). Write a balanced formula equation for this reaction.

19 In addition to acid rain, name two problems that NOx can cause.

Sulfur dioxide (SO_2) All fossil fuels contain some sulfur; sulfur dioxide is formed when fossil fuels burn. Power stations using coal and oil are the major human producers of sulfur dioxide. Volcanoes produce sulfur dioxide. When sulfur dioxide dissolves in water in the atmosphere it forms sulfurous acid (H_2SO_3) which is a contributor to acid rain.

20 Write a balanced formula equation for sulfur dioxide and water forming sulfurous acid.

21 Name a gas (apart from sulfur dioxide) that is a contributor to acid rain.

22 Briefly explain what acid rain is and how it is created/formed.

Particulate matter This term covers all of the tiny particles that can be suspended in the air for long periods of time. Dust, droplets of liquids, soot (carbon particles) and smoke are the major components of particulate matter in the air.

Ozone When this gas is in the upper areas of the atmosphere, it plays an important role in screening out harmful ultra violet (UV) rays. If it is at ground level, ozone irritates the eyes and damages the respiratory system.

23 Which element makes up ozone?

24 Outline one positive contribution of ozone.

25 Outline one negative contribution of ozone.

Research

26 Chlorofluorocarbons (CFCs) are considered a pollutant. Find out what they are used for and what harm they do.

27 Lead used to be added to petrol. Find out why it was used and why this practice has been discontinued.

28 *Pollution will always form a part of our lives; it is how we deal with pollution that matters.* Discuss.

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